

Chapter 4: Loops, Files, and Random Numbers - Multiple Choice Questions

1. What is the primary function of a while loop?

- a) To execute a block of code a specific number of times
- b) To repeat a statement or set of statements as long as a Boolean expression is true
- c) To iterate over a collection of items
- d) To execute a block of code at least once before checking a condition

Answer: b) To repeat a statement or set of statements as long as a Boolean expression is true

2. In a while loop, where is the Boolean expression tested?

- a) At the end of each iteration
- b) In the middle of the loop body
- c) Before each iteration
- d) Only once, before the loop starts

Answer: c) Before each iteration

3. What type of loop is a while loop?

- a) Posttest loop
- b) Pretest loop
- c) Midtest loop
- d) Infinite loop

Answer: b) Pretest loop

4. In the While Loop Demo (Example 1), how many times is the message box displayed?

- a) 0 times
- b) 1 time

- c) 5 times
- d) An infinite number of times

Answer: c) 5 times

5. In the code for Example 1, what is the purpose of the line `count = count + 1`;

- a) To display the current count in the message box
- b) To initialize the loop counter
- c) To prevent an infinite loop by updating the counter
- d) To close the message box

Answer: c) To prevent an infinite loop by updating the counter

6. What will happen if the loop counter in a pretest loop is initialized to a value that makes the Boolean expression false from the start?

- a) The loop will execute once
- b) The loop will execute an infinite number of times
- c) The loop will never execute
- d) The loop will cause a compiler error

Answer: c) The loop will never execute

7. In the Account Balance Application (Example 2), how is the monthly interest applied?

- a) Simple interest calculated once at the end
- b) Compounded monthly
- c) Compounded annually
- d) It is not compounded

Answer: b) Compounded monthly

8. In the enhanced Account Balance Application (Example 3), what control is added to display each month's ending balance?

- a) TextBox
- b) Label

- c) ListBox
- d) Message Box

Answer: c) ListBox

9. Which of the following is NOT one of the three actions specified when writing a for loop?

- a) Initialization
- b) Test
- c) Iteration
- d) Update

Answer: c) Iteration

10. In a for loop header, for (number = 1; number <= 10; number++), what does number++ accomplish?

- a) It tests the loop condition
- b) It initializes the loop counter
- c) It updates the loop counter by incrementing it
- d) It terminates the loop

Answer: c) It updates the loop counter by incrementing it

11. In the Squares application (Example 4), what is the output for the square of 4?

- a) 15
- b) 16
- c) 8
- d) 12

Answer: b) 16 (Note: The PDF had a typo showing 15, but the correct square is 16)

12. What is the scope of a counter variable declared within the initialization expression of a for loop?

- a) It is accessible throughout the entire program
- b) It is accessible only within the loop body

- c) It is accessible only after the loop has finished
- d) It is a global variable

Answer: b) It is accessible only within the loop body

13. In the Speed Converter application (Example 5), what is the conversion factor from KPH to MPH?

- a) 0.6214
- b) 1.6093
- c) 0.5000
- d) 0.7500

Answer: a) 0.6214

14. What is the key characteristic of a do-while loop?

- a) It is a pretest loop
- b) It tests its condition before performing an iteration
- c) It is a posttest loop
- d) It can only be used for reading files

Answer: c) It is a posttest loop

15. Why is it necessary to use files for data storage in many applications?

- a) To make the program run faster
- b) To allow data to be retained after the program stops running
- c) To reduce the amount of code needed
- d) To make the user interface more attractive

Answer: b) To allow data to be retained after the program stops running

16. What is the process of retrieving data from a file called?

- a) Writing data
- b) Saving data

- c) Reading data
- d) Appending data

Answer: c) Reading data

17. What are the three steps that must be taken when a file is used by a program?

- a) Create, Modify, Delete
- b) Read, Write, Execute
- c) Open, Process, Close
- d) Input, Output, Append

Answer: c) Open, Process, Close

18. A file that contains data encoded as text, viewable in an editor like Notepad, is called a:

- a) Binary file
- b) Text file
- c) Random file
- d) Executable file

Answer: b) Text file

19. Which file access method allows you to jump directly to any piece of data without reading the preceding data?

- a) Sequential access
- b) Text access
- c) Direct access
- d) Linear access

Answer: c) Direct access

20. What is an object associated with a specific file that provides a way for a program to work with that file?

- a) A variable
- b) A file object

- c) A control
- d) A namespace

Answer: b) A file object

21. Which .NET Framework class is used to write data to a text file?

- a) StreamReader
- b) FileWriter
- c) StreamWriter
- d) TextWriter

Answer: c) StreamWriter

22. Which namespace must be included in your program to use the StreamWriter and StreamReader classes?

- a) System.Windows
- b) System.Data
- c) System.IO
- d) System.File

Answer: c) System.IO

23. What does the File.CreateText method do if the specified file already exists?

- a) It appends new data to the existing file
- b) It opens the file for reading only
- c) It erases the existing file and creates a new empty one
- d) It throws an exception

Answer: c) It erases the existing file and creates a new empty one

24. What is the key difference between the WriteLine and Write methods of the StreamWriter class?

- a) WriteLine can only write numbers
- b) Write writes a newline character after the data, while WriteLine does not

- c) WriteLine writes a newline character after the data, while Write does not
- d) There is no difference

Answer: c) WriteLine writes a newline character after the data, while Write does not

25. Why is it a good practice to use a try-catch block when performing file operations?

- a) To make the code run faster
- b) To handle potential exceptions, like insufficient rights or a full disk
- c) To automatically close the file
- d) To convert text files to binary files

Answer: b) To handle potential exceptions, like insufficient rights or a full disk

26. In the Friend File application (Example 6), what is written to the file when the user clicks the "Write Name" button?

- a) The current date and time
- b) The contents of the nameTextBox
- c) A random number
- d) The text "Friend File"

Answer: b) The contents of the nameTextBox

27. By default, where is the Friend.txt file created when using File.CreateText("Friend.txt")?

- a) C:\
- b) The project's bin\Debug folder
- c) The user's Documents folder
- d) The desktop

Answer: b) The project's bin\Debug folder

28. When writing numeric data to a text file using WriteLine, what happens to the number?

- a) It is written in binary format
- b) It is converted to a string
- c) It causes an error
- d) It is encrypted

Answer: b) It is converted to a string

29. Which method is used to open a file and append data to its existing contents?

- a) File.CreateText
- b) File.AppendText
- c) File.OpenText
- d) File.AddText

Answer: b) File.AppendText

30. In Example 7, after appending "Lynn", "Steve", and "Bill" to Names.txt, what is the final content?

- a) Joe, Chris, Geni
- b) Lynn, Steve, Bill
- c) Joe, Chris, Geni, Lynn, Steve, Bill
- d) The file is erased and only contains Lynn, Steve, Bill

Answer: c) Joe, Chris, Geni, Lynn, Steve, Bill

31. In the modified Friend File application (Example 8), what is the purpose of nameTextBox.Focus();?

- a) To write the name to the file
- b) To clear the text box
- c) To place the cursor back in the text box for the next entry
- d) To close the form

Answer: c) To place the cursor back in the text box for the next entry

32. How can you specify a different location for an output file, such as C:\Data\file.txt?

- a) Pass the full path as the argument to File.CreateText
- b) It is not possible to change the location
- c) Use the InitialDirectory property of a StreamWriter object
- d) Use the ChangeDirectory method

Answer: a) Pass the full path as the argument to File.CreateText

33. Which .NET Framework class is used to read data from a text file?

- a) StreamWriter
- b) FileReader
- c) StreamReader
- d) TextReader

Answer: c) StreamReader

34. What does the ReadLine method of the StreamReader class do?

- a) Reads the entire file at once
- b) Reads a single line of text from the file and returns it as a string
- c) Reads only numeric data from the file
- d) Writes a line to the file

Answer: b) Reads a single line of text from the file and returns it as a string

35. In Example 9, how many times is MessageBox.Show called?

- a) 1
- b) 4
- c) 5
- d) 0

Answer: b) 4

36. When reading numeric data from a text file, what must you do before performing mathematical operations?

- a) Nothing, you can use the string directly
- b) Convert the string to a numeric data type
- c) Write the number back to the file
- d) Use the ReadNumber method instead of ReadLine

Answer: b) Convert the string to a numeric data type

37. In Example 10, what method is used to convert the strings read from the file into integers?

- a) Convert.ToInteger
- b) int.TryParse
- c) int.Parse
- d) Both b and c are used in the example

Answer: c) int.Parse

38. What property of the StreamReader object indicates whether the end of the file has been reached?

- a) EndOfFile
- b) EOF
- c) EndOfStream
- d) NoMoreData

Answer: c) EndOfStream

39. What is the correct way to write a loop that reads a file until the end is reached?

- a) while (inputFile.EndOfStream == true)
- b) while (!inputFile.EndOfStream)
- c) for (int i=0; i < inputFile.Length; i++)
- d) do {...} while (inputFile.EndOfStream)

Answer: b) while (!inputFile.EndOfStream)

40. In the South America Application (Example 11), what control is used to display the country names?

- a) TextBox
- b) Label
- c) ListBox
- d) ComboBox

Answer: c) ListBox

41. In the Total Sales Application (Example 12), what is the data type of the total variable?

- a) int
- b) double
- c) decimal
- d) string

Answer: c) decimal

42. What is the purpose of the total += sales; statement in Example 12?

- a) To read a sales amount from the file
- b) To display the total sales
- c) To add each sales amount to a running total
- d) To close the file

Answer: c) To add each sales amount to a running total

43. Which control displays a standard Windows "Open" dialog box?

- a) OpenFileDialog
- b) FileDialog
- c) OpenFileDialog
- d) FileOpenDialog

Answer: c) OpenFileDialog

44. Where does an OpenFileDialog control appear when added to a form in the Designer?

- a) On the form itself
- b) In the Toolbox

- c) In the component tray
- d) In the Properties window

Answer: c) In the component tray

45. What value does the ShowDialog method of the OpenFileDialog return if the user clicks the Open button?

- a) DialogResult.Open
- b) DialogResult.OK
- c) DialogResult.Yes
- d) DialogResult.Continue

Answer: b) DialogResult.OK

46. After a user selects a file with the OpenFileDialog, where is the full file path stored?

- a) In the Text property
- b) In the File property
- c) In the Filename property
- d) In the Path property

Answer: c) In the Filename property

47. Which property of the OpenFileDialog control is used to set the initial folder that is displayed?

- a) StartDirectory
- b) InitialFolder
- c) InitialDirectory
- d) FilePath

Answer: c) InitialDirectory

48. Which control is used to display a "Save As" dialog box?

- a) SaveDialog
- b) SaveFileDialog

- c) FileSaveDialog
- d) SaveAsDialog

Answer: b) SaveFileDialog

49. What is the purpose of the Random class in C#?

- a) To generate random numbers
- b) To randomly select files
- c) To put controls in random places on a form
- d) To create random strings

Answer: a) To generate random numbers

50. What is the correct way to create a Random object?

- a) `Random rand = new Random();`
- b) `Random rand = Create.Random();`
- c) `Random rand = Random.New();`
- d) `new Random rand = Random();`

Answer: a) `Random rand = new Random();`

51. If you call `rand.Next(50)`, what is the range of the returned integer?

- a) 1 through 50
- b) 0 through 49
- c) 0 through 50
- d) 1 through 49

Answer: b) 0 through 49

52. How can you generate a random number in the range of 5 through 14?

- a) `rand.Next(5, 14)`
- b) `rand.Next(10) + 5`
- c) `rand.Next(5, 15)`
- d) Both b and c

Answer: d) Both b and c (Note: `rand.Next(5, 15)` generates a number from 5 to 14, and `rand.Next(10) + 5` generates a number from 0+5 to 9+5, which is 5 to 14)

53. What is the range of numbers returned by the `NextDouble` method?

- a) 0.0 through 1.0 (including 1.0)
- b) 0.0 through 1.0 (not including 1.0)
- c) 1.0 through 10.0
- d) -1.0 through 1.0

Answer: b) 0.0 through 1.0 (not including 1.0)

54. In the Coin Toss application (Example 13), if `sideUp` is 0, what is displayed?

- a) Heads
- b) Tails
- c) Both heads and tails
- d) An error message

Answer: b) Tails

55. In the Coin Toss application, what does the line `sideUp = rand.Next(2);` do?

- a) Generates a random number between 1 and 2
- b) Generates a random number between 0 and 1
- c) Generates a random number between 0 and 2
- d) Initializes the Random object

Answer: b) Generates a random number between 0 and 1

56. What event occurs when a form loads into memory?

- a) Click event
- b) Load event
- c) Start event
- d) Initialize event

Answer: b) Load event

57. How do you create a Load event handler for a form in the Designer?

- a) Double-click a button on the form
- b) Double-click the form's title bar
- c) Double-click an empty area of the form
- d) Right-click the form and select "Create Load Event"

Answer: c) Double-click an empty area of the form

58. In a pretest loop, the condition is evaluated _____.

- a) after the loop body executes
- b) before the loop body executes
- c) in the middle of the loop body
- d) only once

Answer: b) before the loop body executes

59. Which loop is guaranteed to execute its body at least once?

- a) for loop
- b) while loop
- c) do-while loop
- d) if statement

Answer: c) do-while loop

60. The File.OpenText method is used to _____.

- a) write data to a new file
- b) append data to an existing file
- c) read data from an existing file
- d) delete a file

Answer: c) read data from an existing file

61. What will the following code snippet do? `outputFile = File.AppendText("data.txt");`

- a) Create a new file named data.txt, erasing any existing one
- b) Open data.txt for reading
- c) Open data.txt for writing, preserving any existing data and appending new data to the end
- d) Delete data.txt

Answer: c) Open data.txt for writing, preserving any existing data and appending new data to the end

62. The `EndOfStream` property is a member of which class?

- a) `StreamWriter`
- b) `File`
- c) `StreamReader`
- d) `Random`

Answer: c) `StreamReader`

63. In the context of file handling, what does "sequential access" mean?

- a) You can jump directly to any piece of data
- b) You must access data from the beginning of the file to the end
- c) Data is accessed in alphabetical order
- d) Data is accessed in reverse order

Answer: b) You must access data from the beginning of the file to the end

64. What is the primary purpose of closing a file after processing?

- a) To delete the file
- b) To save the data (for output files) and free system resources
- c) To make the file read-only
- d) To encrypt the file

Answer: b) To save the data (for output files) and free system resources

65. Which of the following is a valid reason to use a `ListBox` control?

- a) To get single-line text input from the user
- b) To display a list of items from which the user can view or select
- c) To display a single value that the user cannot change
- d) To trigger an event when clicked

Answer: b) To display a list of items from which the user can view or select

66. The Items.Add method is commonly used with which control?

- a) TextBox
- b) Label
- c) ListBox
- d) Button

Answer: c) ListBox

67. What does the TryParse method typically return?

- a) The parsed number
- b) A boolean value indicating success or failure
- c) An exception
- d) A string representation of the number

Answer: b) A boolean value indicating success or failure

68. In the code if (decimal.TryParse(startingBalTextBox.Text, out balance)), what is the purpose of the out keyword?

- a) It specifies that the variable balance is an output parameter
- b) It checks if balance is outside a valid range
- c) It outputs the value of balance to the screen
- d) It declares a new variable

Answer: a) It specifies that the variable balance is an output parameter

69. What is the role of an accumulator variable?

- a) To count the number of loop iterations
- b) To keep a running total or sum of values
- c) To read data from a file
- d) To store the largest value in a set

Answer: b) To keep a running total or sum of values

70. Which loop is best suited when the number of iterations is known beforehand?

- a) while loop
- b) do-while loop
- c) for loop
- d) foreach loop

Answer: c) for loop

71. What is the result of the expression "Result: " + (10 * 2)?

- a) "Result: 20"
- b) "Result: 10*2"
- c) "Result: (10 * 2)"
- d) It causes an error

Answer: a) "Result: 20"

72. The ToString("c") method is used to _____.

- a) format a number as a currency value
- b) format a number as a percentage
- c) convert a string to uppercase
- d) calculate the square root of a number

Answer: a) format a number as a currency value

73. What is the purpose of the Clear method for a ListBox's Items collection?

- a) To clear the selection
- b) To remove all items from the ListBox

- c) To make the ListBox invisible
- d) To reset the ListBox's location

Answer: b) To remove all items from the ListBox

74. In the statement `nameTextBox.Text = ""`;, what is the effect?

- a) It writes an empty string to a file
- b) It clears the contents of the TextBox
- c) It sets the TextBox's background color
- d) It disables the TextBox

Answer: b) It clears the contents of the TextBox

75. What does the Focus method do for a control?

- a) Makes it the active control, ready for user input
- b) Makes it blurry
- c) Hides the control
- d) Increases its font size

Answer: a) Makes it the active control, ready for user input

76. Which property of a PictureBox control determines whether it is visible to the user?

- a) Enabled
- b) Visible
- c) Show
- d) Image

Answer: b) Visible

77. What is the default value of the Visible property for a control?

- a) true
- b) false
- c) null
- d) 0

Answer: a) true

78. The `this.Close();` statement is used to _____.

- a) close the current form and exit the application
- b) close a file
- c) clear all controls on the form
- d) minimize the form

Answer: a) close the current form and exit the application

79. Which of the following is a valid way to generate a random integer between 50 and 100 (inclusive)?

- a) `rand.Next(50, 100)`
- b) `rand.Next(50, 101)`
- c) `rand.Next(49, 100)`
- d) `rand.Next(0, 50) + 50`

Answer: b) `rand.Next(50, 101)`

80. The `Load` event handler executes _____.

- a) when the user clicks a button
- b) when the form is about to be displayed
- c) when the application is compiling
- d) after the form has been closed

Answer: b) when the form is about to be displayed

Wish you all the best!

Fathy.